Scalability in State-Space Search

Two Possible Project Directions

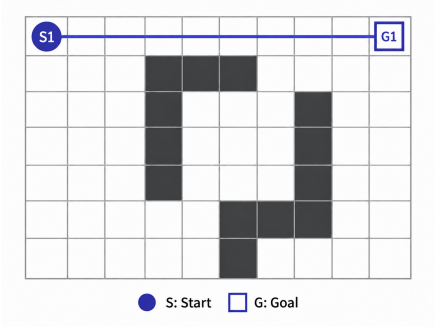
Yihai Li

CIS 5603: Artificial Intelligence



Idea 1: Multi-Agent Path Finding

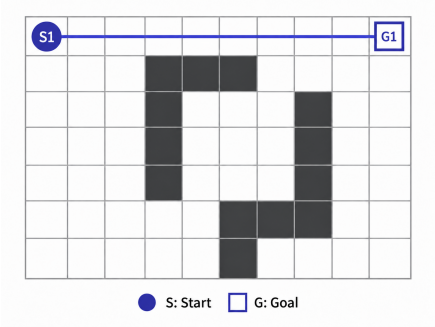
Single-agent path finding



One agent with a start and a goal.

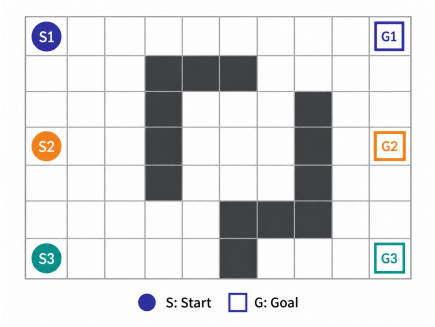
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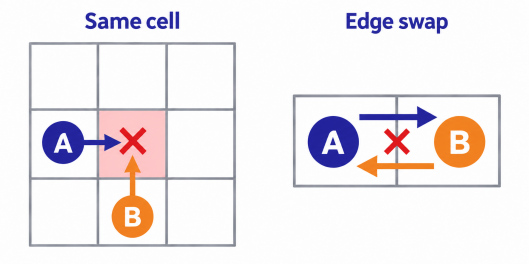
Multi-Agent Path Finding (MAPF)



Several agents share one map.
Each has its own start and goal.
Their paths must avoid conflicts.

MAPF: Conflicts and Scale

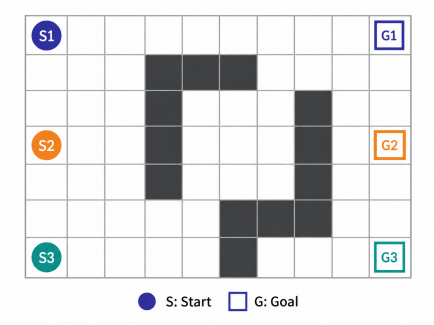
Conflicts between agents



Vertex conflict
Same cell at the same time.

Edge conflict
Opposite moves on the same edge.

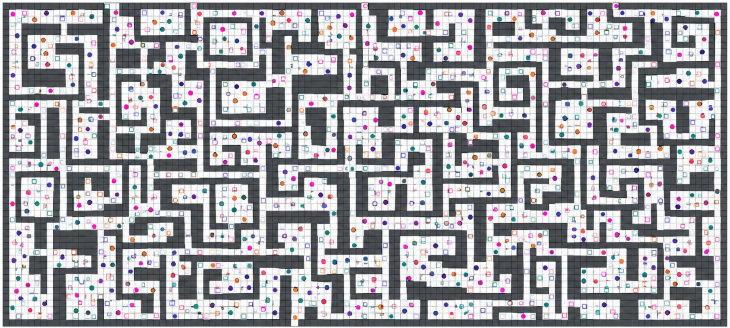
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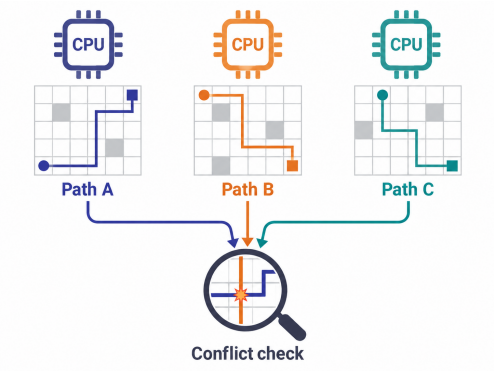
MAPF: Conflicts and Scale

More agents, a larger search space



The search space can grow very quickly.

MAPF: Project Direction

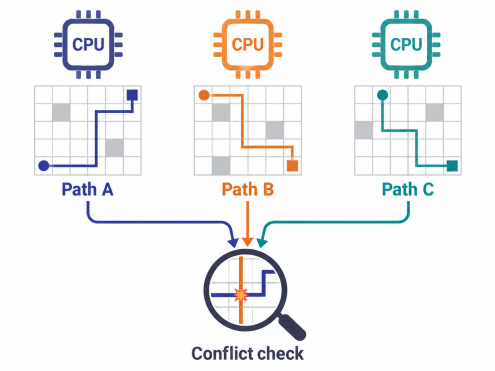


Project focus

Study search-based MAPF.

Identify scalability bottlenecks.

MAPF: Project Direction



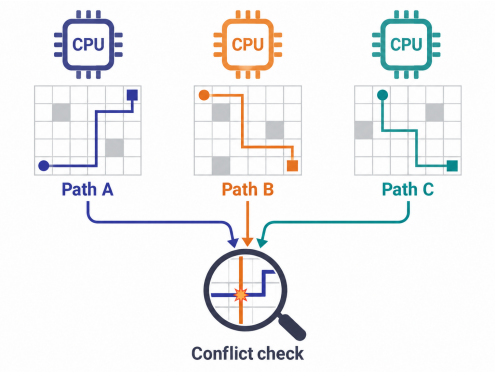
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Possible direction

Parallelize parts of the search.

MAPF: Project Direction



Project focus
Study search-based MAPF.
Identify scalability bottlenecks.

Possible direction
Parallelize parts of the search.

Low-level path searches
Search for different agents' paths simultaneously, when the jobs are independent.

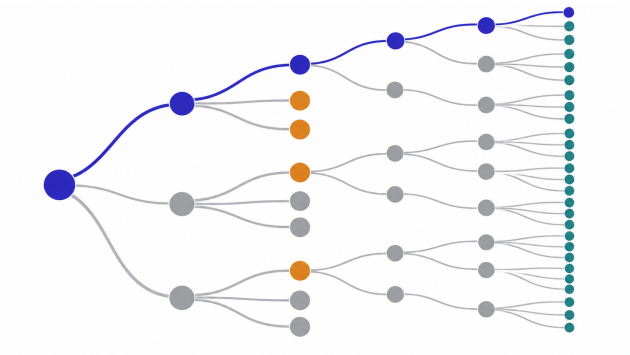
MAPF: Research Question

How can we make search-based

Multi-Agent Path Finding

more scalable?

Idea 2: Parallel Single-Agent Search

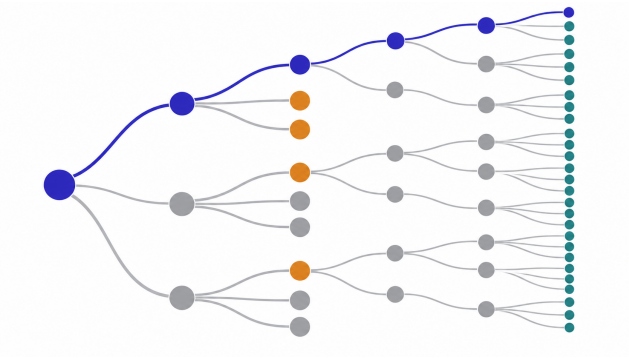


Search states and possible successors.

Sequential BFS or A*

1. Select a state.
2. Expand it.
3. Generate new states.
4. Continue the search.

Idea 2: Parallel Single-Agent Search



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Sequential BFS or A*

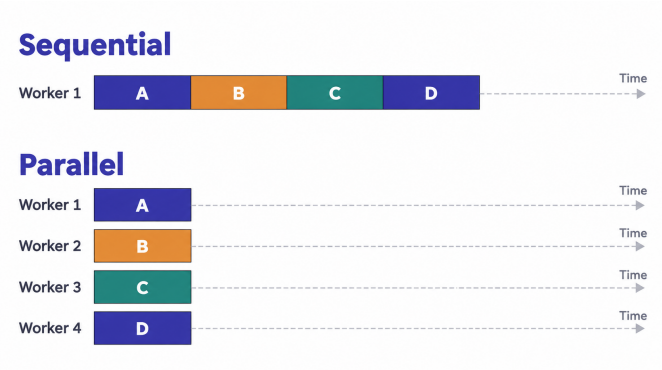
1. Select a state.
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Computational bottlenecks

Large state spaces.

Expensive heuristic evaluations.

Parallel Search: Project Direction

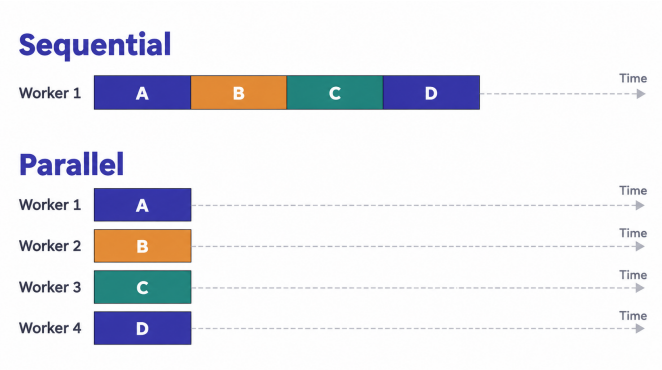


Illustrative task schedule.

Parallel search

Divide search work across workers.

Parallel Search: Project Direction

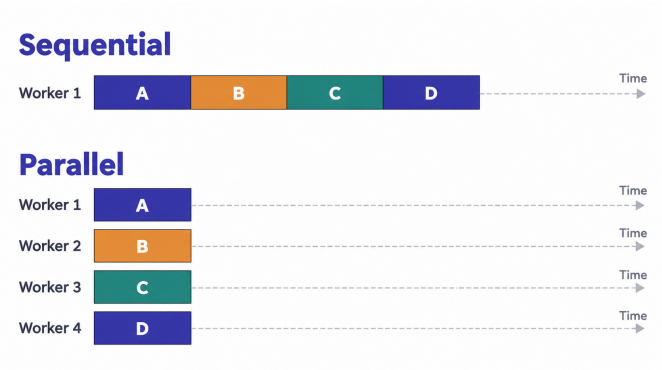


Illustrative task schedule.

Parallel search
Divide search work across workers.

Node expansion
Expand multiple nodes
at the same time.

Parallel Search: Project Direction



Illustrative task schedule.

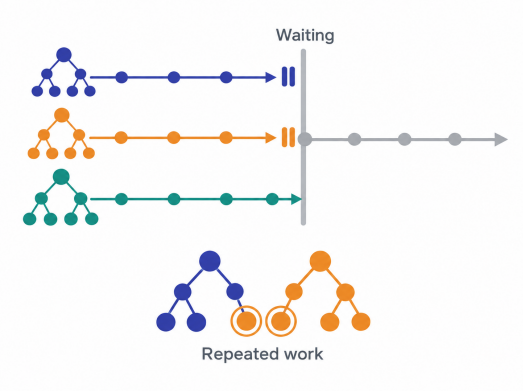
Parallel search
Divide search work across workers.

Node expansion
Expand multiple nodes
at the same time.

Heuristic evaluation
Evaluate multiple states
at the same time.

Comparison
Sequential versus parallel.

Parallel Search Overhead



Parallel search may add work

Search order can change.

Additional states increase work.

Workers may wait to synchronize.

Parallel execution does not always mean faster search.

Parallel Search: Research Question

When does parallelism

improve performance,

and what are the trade-offs?

Project Directions and Scope

MAPF

Application-oriented.

Coordination among multiple agents.

Project Directions and Scope

MAPF

Application-oriented.

Coordination among multiple agents.

Parallel single-agent search

Focus on the search algorithm.

Performance optimization.

Next step

Narrow down the problem and choose a direction
with an appropriate scope for the final project.

Thank you!

Yihai Li

